RevIN Experiment 2: Normalized Backpropagation

1 Theoretical overview and goal

With instance statistics (μ_x, σ_x), data-space MSE is $\|\hat{y} - y\|^2$, whereas normalized backpropagation minimizes

$$\mathrm{nMSE}(\hat{y}, y \mid x) = H^{-1} \sum_{h=1}^{H} \left(\frac{\hat{y}_h - y_h}{\sigma_x + \varepsilon} \right)^2 .$$

The latter gives comparable gradient weight to low- and high-scale windows. Predictions are still denormalized and evaluated with ordinary data-space MSE, so only optimization weighting changes.

Experimental goal. Test whether scale-balanced gradients improve generalization independently of the forward normalization layer, and whether RevIN's affine parameters alter that effect.

2 Implementation details

Eight paired contrasts are run:

Forward transform	Data-space loss	Normalized loss
None	`none_mse`	`none_nmse`
Global standardization	`standard_mse`	`standard_nmse`
Mean only	`mean_only_mse`	`mean_only_nmse`
Scale only	`scale_only_mse`	`scale_only_nmse`
Non-affine instance	`instance_mse`	`instance_nmse`
Median–MAD	`median_mad_mse`	`median_mad_nmse`
Affine RevIN	`revin_mse`	`revin_nmse`
Global MIN	`min_mse`	`min_nmse`

Architecture, seed, batch composition, learning rate, and epoch budget are held fixed within every pair. The paper comparison focuses on MSE versus nMSE.

The loader merges shared, RevIN-scoped, and run-specific user exclusions and records them in every seed directory. This does not replace the separate constant-window audit required by nMSE.

Loss implementation: `src/utils/pipeline.py`
Experiment orchestration: `src/scripts/experiment.py`

The root `revin.slurm` front runs signature-aware training and table stages; resubmission skips current seed and table artifacts.

Small covers five datasets, the three shared default settings, PatchTST, and 16 component methods; full adds all four original ETT panels and two settings, for nine datasets and five settings. Ultra adds DLinear. Publication runs use three seeds and 10,000 steps, validating every 1,000 steps. Tables report data-space test MSE and seed deviations. Validation selection and the optimistic test oracle use every scheduled method by default.

3 Interpretation

nMSE can improve data-space MSE by balancing heterogeneous scales. MSE and nMSE training curves have different units; select with a common validation metric.